

Use the triangle shown for questions 1–2.

1. Write an equation that can be used to find the value of x .

$$90 + (4x + 6) + (6x - 21) = 180$$

OR

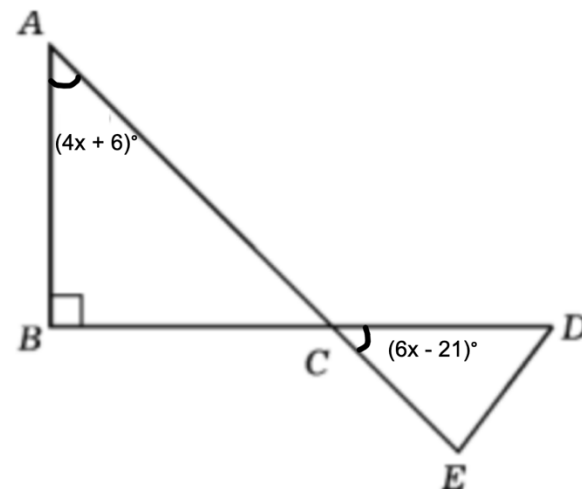
$$(4x + 6) + (6x - 21) = 90$$

2. Solve the equation.

$$10x - 15 = 90$$

$$10x = 105$$

$$x = 10.5$$



Use the diagram for questions 3–5.

3. Write an equation to find the value of d .

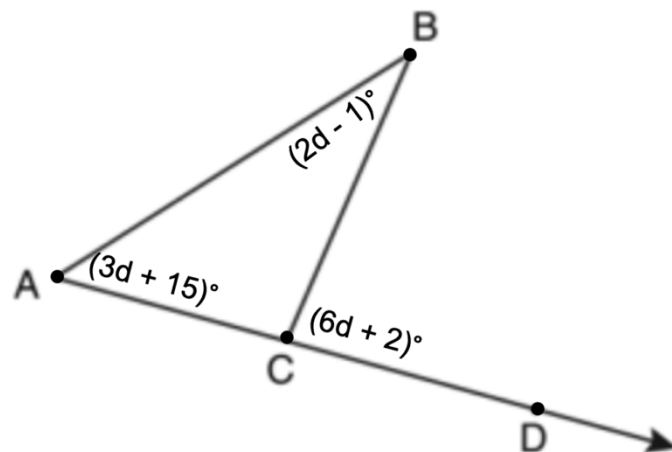
$$(3d + 15) + (2d - 1) = 6d + 2$$

4. Solve the equation.

$$(3d + 15) + (2d - 1) = 6d + 2$$

$$5d + 14 = 6d + 2$$

$$12 = d$$



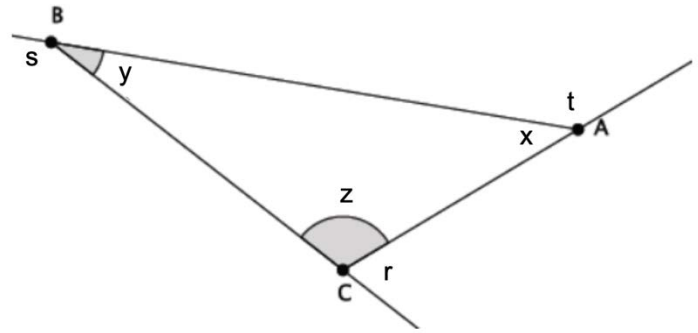
5. Use the solution to find the measure of each angle.

Angle	Measure
$\angle ABC$	$2(12) - 1 = 23^\circ$
$\angle BAC$	$3(12) + 15 = 51^\circ$
$\angle BCD$	$6(12) + 2 = 74^\circ$
$\angle ACB$	$180 - 74 = 106^\circ$

Use the diagram for questions 6–7.

6. Which angle relationship is used to write the equation $z + r = 180$?

- ☐ Exterior Angle Theorem
- ☐ Triangle Sum Theorem
- ☒ Supplementary Angles



7. Which angle relationship is used to write the equation $t = z + y$?

- ☒ Exterior Angle Theorem
- ☐ Triangle Sum Theorem
- ☐ Supplementary Angles

Use the diagram for questions 8–10.

8. Write and solve an equation to find the value of x .

$$\begin{aligned}(10x + 35) + (5x - 11) + 33 &= 180 \\ 15x + 57 &= 180 \\ 15x &= 123 \\ x &= 8.2\end{aligned}$$

9. Write and solve an equation to find the value of y .

$$\begin{aligned}(8y + 1) + (3y + 17) &= 10(8.2) + 35 \\ 11y + 18 &= 117 \\ 11y &= 99 \\ y &= 9\end{aligned}$$

10. Find the measures of the following angles.

$$\angle QSP = \underline{30^\circ}$$

$$5(8.2) - 11 = 30^\circ$$

$$\angle RQS = \underline{63^\circ}$$

$$180 - (10(8.2) + 35) = 63^\circ$$

$$\angle SRT = \underline{107^\circ}$$

$$180 - (8(9) + 1) = 107^\circ$$

